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Lecture 1
Introduction

1.1. Charles Babbage (1791-1871)
Creator of the Analytical Engine - the first general-purpose digital computer (1833)
The Analytical Engine was not built until 1943 (in the form of the Harvard Mark I)

1.2. The Analytical Engine
A programmable, mechanical, digital machine
Could carryout any calculation
Could make decisions based upon the results of the previous calculation
Components: input; memory; processor; output

1.3. Ada, Countess of Lovelace(1815-52)
Babbage: the father of computing
Ada: the mother?
Wrote a program for computing the Bernoulli's sequence on the Analytical Engine - world's 1st computer program
Ada: A programming language specifically designed by the US Dept of Defense for developing military applications was named Ada to honor her contributions towards computing

A lesson that we all can learn from Babbage's Life
Charles Babbage had huge difficulties raising money to fund his research
As a last resort, he designed a clever mathematical scheme along with Ada, the Countess of Lovelace
It was designed to increase their odds while gambling. They bet money on horse races to raise enough money to support their research experiments
Guess what happened at the end? The lost every penny that they had.
Fast
Bored
Storage
Here is a fact:

It could analyze up to 300 billion chess moves in three minutes

In 1997 Deep Blue, a supercomputer designed by IBM, beat Gary Kasparov, the World Chess Champion
That computer was exceptionally fast, did not get tired or bored. It just kept on **analyzing** the situation and kept on **searching** until it found the perfect move from its list of possible moves …

Goals for Today:
To develop an appreciation about the capabilities of computing
To find about the structure & policies of this course

CS101 Introduction to Computing

1.4 Course Contents & Structure

Course Objectives

To build an appreciation for the fundamental concepts in computing
To achieve a beginners proficiency in Web page development
To become familiar with popular PC productivity software

Week	Lecture 1	Lecture 2	Lecture 3 Web Dev	Readings UC	Readings JS	Assignment
1						
2						
3						
4						
5						
6						
7						
8	Midterm Exam					
9						
10						
11						
12						
13						
14						
15						
	Finals Week					

<u>Fundamental concepts</u>

Intro to computing
Evolution of computing
Computer organization
Building a PC
Microprocessors
Binary numbers & logic
Computer software
Operating systems
Application software
Algorithms
Flowcharts
Programming languages

Development methodology
Design heuristics
Web design for usability
Computer networks
Intro to the Internet
Internet services
Graphics & animation
Intelligent systems
Data management
Cyber crime
Social implications
The computing profession
The future of computing

Web page development

Web Development

The World Wide Web
Making a Web page
Lists & tables
Interactive forms
Objective & methods
Data types & operators
Flow control & loops Arrays
Built-in functions
User-defined functions
Events handling
String manipulation
Images & graphics
Programming methodology

Productivity Applications

Word processor
Spreadsheet
Presentation software
Database

Instructor:

Altaf Khan
cs101@vu.edu.pk

Course Web Page:

http://vulms.vu.edu.pk/

Textbooks:

UC - Understanding Computers (2000 ed.)
JS - Learn JavaScript in a Weekend

Reading Assignments

Please make sure to read the assigned material for each week before the commencement of the corresponding week

Reading that material beforehand will help you greatly in absorbing with ease the matter discussed during the lecture
Check your e-mail often for announcements related to this and other VU courses

Marks
distribution …

Assignments (15%)

Almost one every week, 13 in all
No credit for late submissions
The lowest 2 assignment grades will be dropped

Midterm Exam (35%)

During the 8th week
Duration: One hour
Will cover all material covered during the first seven weeks

Final Exam (50%)

During the 16th week
Will cover the whole of the course with a slight emphasis on the material covered after the midterm exam
Duration: 2 hours

First Assignment

Send an email message to me at altaf@vu.edu.pk with the subject "Assignment 1" giving me some information (in around 50 words) about what you see yourself doing ten years from now

Go to the CS101 message board and post a message (consisting of approx. 50 words) about how we could make the contents of this course more suitable for your individual needs. The subject for this message should be "Assignment 1"

Consult the CS101 syllabus for the submission deadline

A suggestion about unfamiliar terms

We try not to use any new terms without explaining them first

However, it is not possible to do that all the time

If you encounter any unfamiliar terms during the lectures, please note them down and consult the GLOSSARY provided at the end of the "Understanding Computers" text book for their meaning

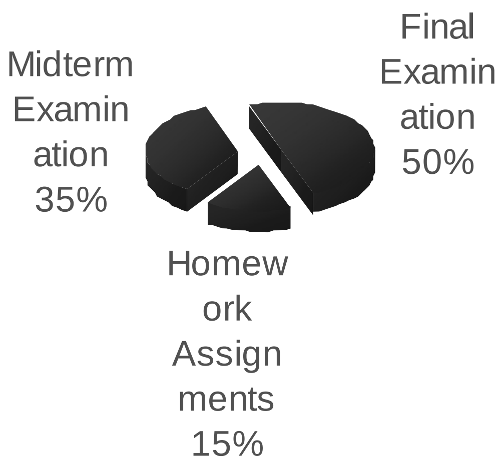
Let's summarize the things that we have covered today?

A few things about:

the very first digital computer & its inventor

the capability of modern computers

the structure and contents of CS101



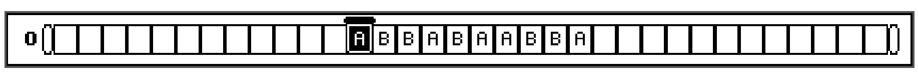
In the Next Lecture …

We'll continue the story of the evolution of digital computers form the Analytical Engine onwards.

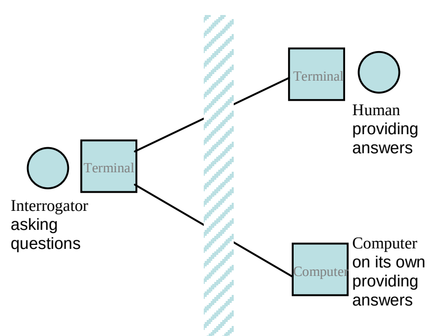
We'll discuss many of the key inventions and developments that he lead to the shape of the current field of computing.

Lecture 2
Evolution of Computing
Today's Goal
To learn about the evolution of computing
To recount the important and key events
To identify some of the milestones in computer development
Babbage's Analytical Engine - 1833
Mechanical, digital, general-purpose
Was crank-driven
Could store instructions
Could perform mathematical calculations
Had the ability to print
Could punched cards as permanent memory
Invented by Joseph-Marie Jacquard

2.1 Turing Machine – 1936
Introduced by Alan Turing in 1936, Turing machines are one of the key abstractions used in modern computability theory, the study of what computers can and cannot do. A Turing machine is a particularly simple kind of computer, one whose operations are limited to reading and writing symbols on a tape, or moving along the tape to the left or right. The tape is marked off into squares, each of which can be filled with at most one symbol. At any given point in its operation, the Turing machine can only read or write on one of these squares, the square located directly below its "read/write" head.



2.2 The "Turing test"
A test proposed to determine if a computer has the ability to think. In 1950, Alan Turing (Turing, 1950) proposed a method for determining if machines can think. This method is known as The Turing Test.
The test is conducted with two people and a machine. One person plays the role of an interrogator and is in a separate room from the machine and the other person. The interrogator only knows the person and machine as A and B. The interrogator does not know which the person is and which the machine is.
Using a teletype, the interrogator, can ask A and B any question he/she wishes. The aim of the interrogator is to determine which the person is and which the machine is. The aim of the machine is to fool the interrogator into thinking that it is a person. If the machine succeeds then we can conclude that machines can think.



2.3 Vacuum Tube – 1904:

A vacuum tube is just that: a glass tube surrounding a vacuum (an area from which all gases has been removed). What makes it interesting is that when electrical contacts are put on the ends, you can get a current to flow though that vacuum.
A British scientist named John A. Fleming made a vacuum tube known today as a diode. Then the diode was known as a "valve,"

2.4 ABC – 1939

The Atanasoff-Berry Computer was the world's first electronic digital computer. It was built by John Vincent Atanasoff and Clifford Berry at Iowa State University during 1937-42. It incorporated several major innovations in computing including the use of binary arithmetic, regenerative memory, parallel processing, and separation of memory and computing functions.

2.5 Harvard Mark 1 – 1943:

Howard Aiken and Grace Hopper designed the MARK series of computers at Harvard University. The MARK series of computers began with the Mark I in 1944. Imagine a giant roomful of noisy, clicking metal parts, 55 feet long and 8 feet high. The 5-ton device contained almost 760,000 separate pieces. Used by the US Navy for gunnery and ballistic calculations, the Mark I was in operation until 1959.
The computer, controlled by pre-punched paper tape, could carry out addition, subtraction, multiplication, division and reference to previous results. It had special subroutines for logarithms and trigonometric functions and used 23 decimal place numbers. Data was stored and counted mechanically using 3000 decimal storage wheels, 1400 rotary dial switches, and 500 miles of wire. Its electromagnetic relays classified the machine as a relay computer. All output was displayed on an electric typewriter. By today's standards, the Mark I was slow, requiring 3-5 seconds for a multiplication operation

2.6 ENIAC – 1946:

ENIAC I (**E**lectrical **N**umerical **I**ntegrator **A**nd **C**alculator). The U.S. military sponsored their research; they needed a calculating device for writing artillery-firing tables (the settings used for different weapons under varied conditions for target accuracy).
John Mauchly was the chief consultant and J Presper Eckert was the chief engineer. Eckert was a graduate student studying at the Moore School when he met John Mauchly in 1943. It took the team about one year to design the ENIAC and 18 months and 500,000 tax dollars to build it.
The ENIAC contained 17,468 vacuum tubes, along with 70,000 resistors and 10,000 capacitors.

2.7 Transistor – 1947

The first transistor was invented at Bell Laboratories on December 16, 1947 by William Shockley. This was perhaps the most important electronics event of the 20th century, as it later made possible the integrated circuit and microprocessor that are the basis of modern electronics. Prior to the transistor the only alternative to its current regulation and switching functions (TRANSfer resISTOR) was the vacuum tubes, which could only be miniaturized to a certain extent, and wasted a lot of energy in the form of heat.
Compared to vacuum tubes, it offered:
smaller size
better reliability
lower power consumption
lower cost

2.8 Floppy Disk – 1950

Invented at the Imperial University in Tokyo by Yoshiro Nakamats

2.9 UNIVAC 1 – 1951

UNIVAC-1. The first commercially successful electronic computer, UNIVAC I, was also the first general purpose computer - designed to handle both numeric and textual information. It was designed by J. Presper Eckert and John Mauchly. The implementation of this machine marked the real beginning of the computer era.

Remington Rand delivered the first UNIVAC machine to the U.S. Bureau of Census in 1951. This machine used magnetic tape for input.
first successful commercial computer
design was derived from the ENIAC (same developers)
first client = U.S. Bureau of the Census
$1 million
48 systems built

2.10 Compiler - 1952

Grace Murray Hopper an employee of Remington-Rand worked on the NUIVAC. She took up the concept of reusable software in her 1952 paper entitled "The Education of a Computer" and developed the first software that could translate symbols of higher computer languages into machine language. (Compiler)

2.11 ARPANET – 1969

The Advanced Research Projects Agency was formed with an emphasis towards research, and thus was not oriented only to a military product. The formation of this agency was part of the U.S. reaction to the then Soviet Union's launch of Sputnik in 1957. (ARPA draft, III-6). ARPA was assigned to research how to utilize their investment in computers via Command and Control Research (CCR). Dr. J.C.R. Licklider was chosen to head this effort.
Developed for the US DoD Advanced Research Projects Agency
60,000 computers connected for communication among research organizations and universities

2.12 Intel 4004 – 1971

The 4004 was the world's first universal microprocessor. In the late 1960s, many scientists had discussed the possibility of a computer on a chip, but nearly everyone felt that integrated circuit technology was not yet ready to support such a chip. Intel's Ted Hoff felt differently; he was the first person to recognize that the new silicon-gated MOS technology might make a single-chip CPU (central processing unit) possible.
Hoff and the Intel team developed such architecture with just over 2,300 transistors in an area of only 3 by 4 millimeters. With its 4-bit CPU, command register, decoder, decoding control, control monitoring of machine commands and interim register, the 4004 was one heck of a little invention. Today's 64-bit microprocessors are still based on similar designs, and the microprocessor is still the most complex mass-produced product ever with more than 5.5 million transistors performing hundreds of millions of calculations each second - numbers that are sure to be outdated fast.

2.13 Altair 8800 – 1975

By 1975 the market for the personal computer was demanding a product that did not require an electrical engineering background and thus the first mass produced and marketed personal computer (available both as a kit or assembled) was welcomed with open arms. Developers Edward Roberts, William Yates and Jim Bybee spent 1973-1974 to develop the MITS (Micro Instruments Telemetry Systems) Altair 8800. The price was $375, contained 256 bytes of memory (not 256k),but had no keyboard, no display, and no auxiliary storage device. Later, Bill Gates and Paul Allen wrote their first product for the Altair -- a BASIC compiler (named after a planet on a *Star Trek* episode).